PROPERNESS OF THE SCHEME THEORETIC SUPPORT APPEARS AT A FINITE STAGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3238-3253, following the statement at 3223-3236.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Supports commute with pullback. Morphisms of Spaces, Lemma spaces-morphisms-lemma-support-finite-type (cross-chapter, cited at ~3240-3242). Replacing X_0 by the support of F_0 makes X_i and X the supports of F_i and F.
- Image of a proper morphism is proper. Morphisms, Lemma morphisms-lemma-image-proper-is-proper (cross-chapter, cited at ~3246-3248). Z -> Y proper plus universal homeomorphism Z -> X gives X -> Y proper.
- Eventual properness and properness lemmas. lemma-eventually-proper (same chapter, cited at ~3250-3251); Morphisms of Spaces, Lemmas spaces-morphisms-lemma-closed-immersion-proper and spaces-morphisms-lemma-composition-proper (cross-chapter, cited at ~3251-3253). X_i -> Y proper for some i; Z_i -> X_i closed immersion gives Z_i proper by composition.

Source reference. Original lemma 08K1: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2023–2023. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03GN: focused-seeds/geometry/S9-spaces-limits-eventually-proper-support/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04XZ: focused-seeds/geometry/S9-spaces-limits-eventually-proper-support/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04XY: focused-seeds/geometry/S9-spaces-limits-eventually-proper-support/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07TZ: focused-seeds/geometry/S9-spaces-limits-eventually-proper-support/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 084W. Let $\mathcal{F}_0$ be a quasi-coherent $\mathcal{O}_{X_0}$-module. Denote $\mathcal{F}$ and $\mathcal{F}_i$ the pullbacks of $\mathcal{F}_0$ to X and X_i. Assume*

(1) *f_0 is locally of finite type,*
(2) *$\mathcal{F}_0$ is of finite type,*
(3) *the scheme theoretic support of $\mathcal{F}$ is proper over Y.*

Then the scheme theoretic support of $\mathcal{F}_i$ is proper over Y_i for some i.

Proof. We may replace X_0 by the scheme theoretic support of $\mathcal{F}_0$. By Morphisms of Spaces, Lemma 07TZ this guarantees that X_i is the support of $\mathcal{F}_i$ and X is the support of $\mathcal{F}$. Then, if $Z \subset X$ denotes the scheme theoretic support of $\mathcal{F}$, we see that $Z \to X$ is a universal homeomorphism. We conclude that $X \to Y$ is proper as this is true for $Z \to Y$ by assumption, see Morphisms, Lemma 03GN. By Lemma 08K1 we see that $X_i \to Y$ is proper for some i. Then it follows that the scheme theoretic support Z_i of $\mathcal{F}_i$ is proper over Y by Morphisms of Spaces, Lemmas 04XZ and 04XY. □